SREY PATEL

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EDUCATION

Master's of Engineering in System and Technology (Co-op) Hamilton, ON
McMaster University Sep 2023 – May 2025
Coursework: Deep Learning, AI/ML Fundamentals, Cloud Computing, Natural Language Processing, Leadership for Innovation GPA: 3.7 / 4.0

Bachelor's of Engineering in Computer Engineering Ahmedabad, IN
Gujarat Technological University Jul 2019 – May 2023
Relevant Coursework: Software Engineering, Web programming, Analysis and Design of Algorithm, Python for Data Science, Artificial Intelligence, Machine Learning, Data Visualization. GPA: 3.78 / 4.0

SKILLS

Programming Skills: C++, Python3, JavaScript, Java, Android (Java), C, HTML5, CSS3.
Technical Skills: Software Engineering, Object-Oriented Programming, Data structures, Version Control.
Soft Skills: Teamwork**,** Effective Communication, Organizational Skills, Adaptability and Flexibility, Innovative Thinking, Mentoring
Technologies/Frameworks: ReactJS, NodeJS, Django, Bootstrap, TailwindCSS.
Tools/Database: Jupyter Notebook, VSCode, Postman, AWS, Git, GitHub, Postman, Slack, Anaconda, PowerBI, Microsoft Excel, Microsoft Word, Microsoft Powerpoint, MySQL, NoSQL, MongoDB, PostgreSQL.

PROFESSIONAL EXPERIENCE

Junior Software Developer (Co-op) September 2024 – December 2024
Hatch Ltd Niagara, Canada

- Analyzing hydropower inflow datasets to forecast inflow patterns based on key factors such as temperature, precipitation, channel input, rain, and snow melting, enhancing the accuracy of resource management.
- Implementing machine learning models, including LSTM, XGBoost, and Random Forest, to predict inflow trends and optimize energy production forecasts for hydropower dam operations, achieving 93.77% accuracy with the LSTM model.
- Utilized Microsoft Azure for project repository management and cloud service deployment, enhancing data handling and infrastructure support for machine learning and analytics in hydropower operations.

Software Developer Intern February 2023 – May 2023
Grownited Pvt.Ltd Ahmedabad, India

- Collaborated within a dynamic team to architect the platform, delivering an exceptional user experience with seamless, responsive design and peak performance across multiple devices. This effort resulted in a improvement in website performance.
- Engineered server-side solutions with Node.js and mastered backend API management and creation using Postman. These enhancements catapulted the website's performance by 30% and streamlined the overall system's reliability and efficiency.
- Crafted a user-focused interface with React.js, meticulously applying contemporary UI/UX principles. This deliberate redesign achieved a significant 15% boost in user satisfaction, fostering enhanced engagement and deeper loyalty among the platform's customer base.

PROJECTS

AI Teaching Assistent | NLP, LLMA2, Chroma DB, LangChain, PyPdf, PdfMining, ChainLit October 2024

- Developed a Teaching Assistant AI on Chainlit with a RAG backend, extracting and embedding course materials using PyPDF2 and Chroma DB for efficient retrieval.
- Utilized Langchain to build a retrieval chain, fetching the top 3 relevant data points, and applied prompt engineering techniques alongside the LLAMA2 model to generate accurate question-answer responses based on the retrieved data

Computer Based Testing | Python, Django, Mysql February 2024

- Built a feature-rich, mobile-responsive exam platform using Django's MVC architecture, optimizing test workflows, automating grading, and reducing exam duration by 40%.
- Implemented a robust role-based access control system with multi-factor authentication, cutting unauthorized access by 95%, improving security, and streamlining exam administration for various users.

WatchWave | ReactJS, Tailwind CSS, Rapid API June 2023

- Developed WatchWave, a ReactJS-based video streaming platform with Material UI 5, custom categories, and YouTube API integration for a seamless and engaging user experience.
- Ensured optimal user interaction by creating responsive channel and video pages with distinctive video cards, while also deploying the application on Netlify for widespread accessibility and seamless online streaming.

CERTIFICATIONS

- Learning Git and Github: LinkedIn Learning(2024).
- Frontend Development with ReactJS and Bootstrap: Udemy(2023).
- The Complete MySQL Course: Udemy(2022).